Supplementary Material: Prenatal Mendelian Randomization Systematic Review

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I. Details of search terms used in systematic review

embase.com	218	215
Medline Ovid	177	24
Web of science	250	129
Cochrane CENTRAL	16	1
Google scholar	200	136
Total	**861**	**505**

embase.com 218

('instrumental variable analysis'/exp OR ((mendelian* NEAR/3 random*) OR (instrumental* NEAR/3 variab*)):ab,ti) AND ('prenatal exposure'/exp OR 'prenatal drug exposure'/exp OR 'pregnant woman'/exp OR 'pregnancy'/exp OR 'prenatal disorder'/exp OR 'pregnancy disorder'/exp OR 'parameters concerning the fetus, newborn and pregnancy'/exp OR 'prenatal development'/exp OR 'maternal nutrition'/exp OR 'maternal smoking'/exp OR 'Maternal Exposure'/exp OR 'embryonic and placental structures'/exp OR (prenatal* OR perinatal* OR pregnan* OR in*-uter* OR intrauter* OR gestation* OR maternal* OR offspring OR birthweight OR birth-weight OR fetus OR fetal OR foetus OR foetal OR placenta* OR embryo* OR fetomaternal* OR PreEclampsia OR Eclampsia):ab,ti)

Medline Ovid 262

(Mendelian Randomization Analysis/ OR ((mendelian* ADJ3 random*) OR (instrumental* ADJ3 variab*)).ab,ti.) AND (exp Pregnancy Complications/ OR Maternal Exposure/ OR pregnant women/ OR exp pregnancy/ OR exp Fetal Diseases/ OR exp Pregnancy Complications/ OR exp Birth Weight/ OR exp Infant, Low Birth Weight/ OR Perinatal Mortality/ OR Perinatal Death/ OR Embryology/ OR exp "Embryonic and Fetal Development"/ OR exp Maternal Nutritional Physiological Phenomena/ OR exp Embryonic Structures/ OR (prenatal* OR perinatal* OR pregnan* OR in*-uter* OR intrauter* OR gestation* OR maternal* OR offspring OR birthweight OR birth-weight OR fetus OR fetal OR foetus OR foetal OR placenta* OR embryo* OR fetomaternal* OR PreEclampsia OR Eclampsia).ab,ti.)

Cochrane CENTRAL 16

(((mendelian* NEAR/3 random*) OR (instrumental* NEAR/3 variab*)):ab,ti) AND ((prenatal* OR perinatal* OR pregnan* OR in*-uter* OR intrauter* OR gestation* OR maternal* OR offspring OR birthweight OR birth-weight OR fetus OR fetal OR foetus OR foetal OR placenta* OR embryo* OR fetomaternal* OR PreEclampsia OR Eclampsia):ab,ti)

Web of science 250

TS=((((mendelian* NEAR/2 random*) OR (instrumental* NEAR/2 variab*))) AND ((prenatal* OR perinatal* OR pregnan* OR in*-uter* OR intrauter* OR gestation* OR (maternal* NEAR/3 (exposure* OR smoking OR drinking OR alcohol)) OR offspring OR birthweight OR birth-weight OR fetus OR fetal OR foetus OR foetal OR placenta* OR embryo* OR fetomaternal* OR PreEclampsia OR Eclampsia)))

Google scholar

"mendelian randomization|randomisation"|"instrumental variable" prenatal|perinatal|pregnancy|pregnant|"in-uterus"|intrauterine|gestational|maternal|offspring|birthweight|"birth-weight"|fetus|fetal|foetus|foetal|placenta|embryo|fetomaternal

II. Details of extraction procedure

Data points were extracted by the first author (ED); to ensure accuracy in extraction, 5 included studies were randomly chosen for independent extraction by a coauthor (JL). Data points on discussion of MR assumptions were considered in agreement when both authors agreed on the presence/absence of any discussion of violations of the assumption in question.

Sensitivity analyses and limitations were excluded from the comparison checking due to variability in how specific secondary analyses and limitations were categorized. This is because, rather than prespecifying sets of possible limitations and sensitivity analyses of interest, extraction of data points related to both sensitivity analyses and limitations discussed were open-ended to allow for unexpected or unknown analyses and perspectives. This approach meant that each independent extractor could generate an arbitrarily large number of reported limitations and sensitivity analyses based on the same article. Thus, it would be difficult to measure the degree to which independent extractors agreed on datapoints related to sensitivity analyses and limitations discussed, as it is not possible to measure the number of datapoints the two authors agreed were not present in the dataset.

III. Supplementary Table 1: Description of Included Studies

Supplementary Table 1. Description of Included Studies

1: Author	2: Year	3: Title	4: Exposure	5: Outcome	6: Instrument	7: Maternal or Offspring Instrument	8: Design	9: Recruited Based on Presence of a Pregnancy
Allard	2015	Mendelian randomization supports causality between maternal hyperglycemia and epigenetic regulation of leptin gene in newborns	2 step: maternal fasting glucose, methylation	2 step: methylation, cord blood leptin	2 step: glucose GRS, methylation GRS	1st step: maternal	cohort	yes
Alwan	2012	Exploring the relationship between maternal iron status and offspring's blood pressure and adiposity: A Mendelian randomization Study.	iron	blood pressure, waist circumference, BMI	C282Y	maternal	cohort	No
Bech	2006	Stillbirth and slow metabolizers of caffeine: Comparison by genotypes.	caffeine	stillbirth	NAT2, CYP1A2, GSTA1	maternal	nested case-control	yes (nested in recruited that way)

Bedard	2018	Maternal iron status during pregnancy and respiratory and atopic outcomes in the offspring: A Mendelian randomisation study	hemoglobin	wheezing, asthma, atopy, low lung function	maternal 12 SNP weighted GRS	maternal	cohort	Yes
Bernard	2018	Long-chain polyunsaturated fatty acids, gestation duration, and birth size: A Mendelian randomization study using fatty acid desaturase variants	omega 3 and omega 6 PUFAs	gestational duration, birthweight, birth length	8 FADS variants	maternal and offspring	cohort	yes
Binder	2013	The causal effect of red blood cell folate on genome-wide methylation in cord blood: a Mendelian randomization approach	folate	genome-wide methylation	MTHFR rs1801133, rs1801131	maternal	cross-sectional	Yes
Bonilla	2012	Maternal and offspring fasting glucose and type 2 diabetes-associated genetic variants and cognitive function at age 8: A Mendelian randomization study in the Avon Longitudinal Study of Parents and Children	fasting glucose, type 2 diabetes	IQ at age 8	genetic risk score	maternal and offspring	cohort	Yes

Bonilla	2012	Vitamin B-12 Status during Pregnancy and Child's IQ at Age 8: A Mendelian Randomization Study in the Avon Longitudinal Study of Parents and Children	vitamin B12	IQ at age 8	rs492602, rs1801198, rs96-6756	maternal	cohort	Yes
Caramaschi	2017	Exploring a causal role of DNA methylation in the relationship between maternal vitamin B12 during pregnancy and child's IQ at age 8, cognitive performance and educational attainment: a two-step Mendelian randomization study.	2 step: vitamin B12, methylation	2 step: methylation, IQ	2 step: rs492602 + rs1047781 for vitamin b12, rs5750236, rs1890131 for methylation	maternal, offspring in 2nd stpe	cohort	Yes
Caramaschi	2018	Maternal smoking during pregnancy and autism: using causal inference methods in a birth cohort study	smoking heaviness	autism spectrum disorder	rs1051730	maternal	cohort	Yes
Evans	2018	Elucidating the role of maternal environmental exposures on offspring health and disease using two-sample Mendelian randomization	maternal type 2 diabetes	birthweight	403 SNP GRS	maternal and offspring	cross-sectional, UK biobank	No

Geng	2018	Maternal central obesity and birth size: A Mendelian randomization analysis	waist-to-hip ratio adjusted for BMI, hip circumference adjusted for BMI, waist circumference adjusted for BMI	birth weight, birth length, head circumferenc e,	35, 25, and 41 SNP GRS	maternal	2 sample cross-sectional, seems to be same studies that developed GRS in?	No
Granell	2008	The association between mother and child MTHFR C677T polymorphisms, dietary folate intake and childhood atopy in a population-based, longitudinal birth cohort	folate	atopy, asthma	MTHFR C677T	maternal	cohort	yes
Howe	2019	Prenatal alcohol exposure and facial morphology in a UK cohort	alcohol	facial morphology	rs1229984	maternal	cohort	Yes
Humphriss	2013	Prenatal alcohol exposure and childhood balance ability: Findings from a UK birth cohort study	alcohol	3 composite balance scores (dynamic balance, static balance eyes open, static balance eyes closed)	ADH1B rs1229984	maternal	cohort	Yes

Hwang	2019	Using a two-sample Mendelian randomization design to investigate a possible causal effect of maternal lipid concentrations on offspring birth weight	HDL cholesterol, LDL cholesterol, triglycerides	birthweight	96, 82, and 60 SNP GRS	maternal, controlled for offspring using SEM	2 sample summary results, UK biobank and EGG	uK biobank no, EGG mostly yes
Korevaar	2014	Soluble Flt1 and Placental Growth Factor are novel determinants of newborn thyroid dysfunction: the generation r study. J clin Endocrinol Metab	TSH, FT4	sFlt1, PlGF	GRS	offspring	cohort	yes
Lawlor	2008	Exploring the developmental overnutrition hypothesis using parental-offspring associations and FTO as an instrumental variable	BMI	fat mass at age 9-11	FTO	maternal	cohort	Yes
Lawlor	2017	Using Mendelian randomization to determine causal effects of maternal pregnancy (intrauterine) exposures on offspring outcomes: Sources of bias and methods for assessing them	BMI	BMI, fat mass index	GRS	maternal	cohort	Yes

Lee	2013	Mendelian randomization analysis of the effect of maternal homocysteine during pregnancy, as represented by maternal MTHFR C677T genotype, on birth weight	homocysteine	birthweight	MTHFR C677T	maternal	cohort	Yes
Lewis	2009	Body composition at age 9 years, maternal folate intake during pregnancy and methyltetrahydrofolate reductase (MTHFR) C677T genotype	folate intake	total weight,total body fat mass, total lean mass	MTHFR C677T	maternal and offspring	cohort	yes
Lewis	2012	Fetal Alcohol Exposure and IQ at Age 8: Evidence from a Population-Based Birth-Cohort Study	alcohol	cognitive score (IQ at age 8)	10 SNP in ADH4, ADH1A, AHD1B, ADH7 (rs4699714, rs3763894, rs4148884, rs2866151, rs975833, rs1229966, rs2066701, rs4147536, rs1229984, rs284779)	both	cohort	Yes
Lewis	2014	Maternal iron levels early in pregnancy are	iron	IQ at age 8	GRS based on	maternal	cohort	Yes

		not associated with offspring IQ score at age 8, findings from a Mendelian randomization study			rs1799945, rs1800562, rs4820268			
Mamasoula	2013	Association between C677T Polymorphism of Methylene Tetrahydrofolate reductase and Congenital Heart Disease: Meta-Analysis of 7697 Cases and 13,125 Controls. Circ Cardiovasc Genet	folate	congenital heart disease	MTHFR rs1801133	maternal and offspring	case-control	cases yes, controls no
Morales	2011	Maternal C-reactive protein levels in pregnancy are associated with wheezing and lower respiratory tract infections in the offspring	CRP	wheezing, LRTI	rs1205	maternal	cohort	Yes
Morales	2016	Genome-wide DNA methylation study in human placenta identifies novel loci associated with maternal smoking during pregnancy	methylation at top-ranked cpg site for placental methylation in smokers	birthweight	rs1983204, rs344008, rs6795327, rs7637701, rs11929637	offspring	cohort	Yes
Murray	2016	Moderate alcohol drinking in pregnancy increases risk for children's persistent	alcohol	conduct problem trajectories (6 measures of	GRS ADH1A rs2866151, rs975833, AHD1B	offspring	cohort	Yes

		conduct problems: causal effects in a Mendelian randomisation study		strengths and difficulties questionnaire)	rs4147536, ADH7 rs284779			
Richmond	2016	DNA methylation and BMI: Investigating identified methylation sites at HIF3A in a causal framework	BMI	HIF3A methylation	GRS	maternal	cohort	Yes
Richmond	2017	Using Genetic Variation to Explore the Causal Effect of Maternal Pregnancy Adiposity on Future Offspring Adiposity: A Mendelian Randomisation Study	BMI	BMI, fat mass index	GRS	maternal	cohort	yes
von Hinke Kessler Scholder	2014	Alcohol Exposure In Utero and Child Academic Achievement	alcohol	academic achievement (KS1,KS2,KS3, GCSE)	ADH1B rs1229984	maternal, controllin g for offspring	cohort	Yes
Shaheen	2014	Prenatal alcohol exposure and childhood atopic disease: A Mendelian randomization approach	alcohol	childhood atopic disease	ADH1B rs1229984	maternal	cohort	Yes
Steenweg-de Graaff	2012	Maternal folate status in early pregnancy and child emotional and behavioral problems: the Generation R Study	folate	CBCL emotional and behavioral score	MTHFR C677T	maternal	cohort	yes

Steer	2011	Insights into the programming of bone development from the Avon Longitudinal Study of Parents and Children (ALSPAC)	folate	BMC, BMD, BA	MTHFR C677T	maternal	cohort	yes
Taylor	2014	Maternal smoking during pregnancy and offspring smoking initiation: assessing the role of intrauterine exposure	smoking	latent class of offspring smoking initiation	rs1051730	maternal	cohort	yes
Thompson	2019	Association of maternal circulating 25(OH)D and calcium with birth weight: A mendelian randomisation analysis	vitamin D, calcium	birth weight	separate 7 SNP GRS	maternal	2 sample: various GWAS+ UKB cross-sectional, sensitivity analyses: ALSPAC/EFSOCH cohort	ALSPAC/EFSOCH yes, UKB no
Tyrell	2016	Genetic evidence for causal relationships between maternal obesity related traits and birth weight. JAMA	BMI, fasting glucose, diabetes, triglycerids, HDL, blood pressure, vitamin D, adiponectin	birthweight	GRS	maternal	meta-analysis of multiple cohorts	yes for all
Wehby	2011	A genetic instrumental variables analysis of the effects of prenatal smoking on birth weight: Evidence from two samples	smoking	birthweight	14 SNPs	maternal	1 cohort, 1 cross-sectional	yes for Norway, no for AddHealth

Wehby	2011	Genes as instruments for studying risk behavior effects: An application to maternal smoking and orofacial clefts	smoking	orofacial cleft	4 SNPS (rs1435252, rs1930139, rs1547272, rs2743467	maternal	case-control	yes
Wehby	2013	Genetic instrumental variable studies of effects of prenatal risk factors	smoking. Alcohol use, obesity	birthweight	smoking: rs12914385, rs1051730, alcohol: ADH1B rs1229984, BMI: rs8050136	maternal	cohort (multiple)	yes for both
Yajnik	2014	Maternal homocysteine in pregnancy and offspring birthweight: Epidemiological associations and Mendelian randomization analysis	homocysteine	birthweight	MTHFR rs1801133	maternal	2 cohorts	yes
Zerbo	2016	Maternal mid-pregnancy C-reactive protein and risk of autism spectrum disorders: the early markers for autism study	CRP	autism spectrum disorder	rs3116656, rs2794520	maternal	nested case-control	yes
Zhang	2015	Assessing the Causal Relationship of Maternal Height on Birth Size and Gestational Age at	maternal height	birth length, birth weight	GRS	maternal	3 case-control nested w/in cohorts	yes for all

		Birth: A Mendelian Randomization Analysis						
Zuccolo	2013	Prenatal alcohol exposure and offspring cognition and school performance. A 'Mendelian randomization' natural experiment	alcohol (1st trimester)	IQ at age 8, educational attainment	rs1229984	maternal, sensitivity analysis with offspring	cohort	yes

Table 1 Continued

1: Author	2: Year	3: Title	10: Point Estimation?	11: F-statistic	12: Assumption 1 Violations Discussed	13: Assumption 2 Violations Discussed	14: Assumption 3 Violations Discussed
Allard	2015	Mendelian randomization supports causality between maternal hyperglycemia and epigenetic regulation of leptin gene in newborns	point estimate				Population Stratification
Alwan	2012	Exploring the relationship between maternal iron status and offspring's blood pressure and adiposity: A Mendelian randomization Study.	point estimate	10	Weak Instrument Bias	Pleiotropy, Exposure Measurement Error, Offspring Genotype Path	
Bech	2006	Stillbirth and slow metabolizers of caffeine: Comparison by genotypes.	instrument-outcome association		Weak Instrument Bias		
Bedard	2018	Maternal iron status during pregnancy and respiratory and atopic outcomes in the offspring: A Mendelian randomisation study	point estimate			Pleiotropy,Exposure Measurement Error,Offspring Genotype Path	Population Stratification
Bernard	2018	Long-chain polyunsaturated fatty acids, gestation duration, and birth size: A Mendelian randomization study using fatty acid desaturase variants	instrument-outcome association		Weak Instrument Bias	Pleiotropy,Exposure Measurement Error	Population Stratification
Binder	2013	The causal effect of red blood cell folate on genome-wide methylation in cord blood: a Mendelian randomization approach	point estimate	4.876	Weak Instrument Bias	2nd or 3rd Assumption - General	Population Stratification
Bonilla	2012	Maternal and offspring fasting glucose and type 2 diabetes-associated genetic variants and	instrument-outcome association			Offspring Genotype Path	Population Stratification

		cognitive function at age 8: A Mendelian randomization study in the Avon Longitudinal Study of Parents and Children					
Bonilla	2012	Vitamin B-12 Status during Pregnancy and Child's IQ at Age 8: A Mendelian Randomization Study in the Avon Longitudinal Study of Parents and Children	instrument-outcome association			Exposure Measurement Error, Offspring Genotype Path	Population Stratification
Caramaschi	2017	Exploring a causal role of DNA methylation in the relationship between maternal vitamin B12 during pregnancy and child's IQ at age 8, cognitive performance and educational attainment: a two-step Mendelian randomization study.	point estimate			Pleiotropy, Exposure Measurement Error	Population Stratification
Caramaschi	2018	Maternal smoking during pregnancy and autism: using causal inference methods in a birth cohort study	instrument-outcome association			Pleiotropy, Exposure Measurement Error	
Evans	2018	Elucidating the role of maternal environmental exposures on offspring health and disease using two-sample Mendelian randomization	point estimate		Weak Instrument Bias	Pleiotropy, Offspring Genotype Path	Population Stratification
Geng	2018	Maternal central obesity and birth size: A Mendelian randomization analysis	point estimate		Weak Instrument Bias	Pleiotropy, Postnatal Effects of Genotype, Offspring Genotype Path,	Population Stratification
Granell	2008	The association between mother and child MTHFR C677T polymorphisms, dietary folate intake and childhood atopy in a population-based, longitudinal birth cohort	instrument-outcome association		Weak Instrument Bias		

Howe	2019	Prenatal alcohol exposure and facial morphology in a UK cohort	instrument-outcome association			Pleiotropy, Offspring Genotype Path	Assortative Mating
Humphriss	2013	Prenatal alcohol exposure and childhood balance ability: Findings from a UK birth cohort study	instrument-outcome association		Weak Instrument Bias, Cannot Prove Assumption 1 (Reduced Form)	Pleiotropy	Population Stratification
Hwang	2019	Using a two-sample Mendelian randomization design to investigate a possible causal effect of maternal lipid concentrations on offspring birth weight	point estimate			Pleiotropy, Postnatal Effects of Genotype, Exposure Assumed Constant Over Pregnancy, Offspring Genotype Path	Population Stratification
Korevaar	2014	Soluble Flt1 and Placental Growth Factor are novel determinants of newborn thyroid dysfunction: the generation r study. J clin Endocrinol Metab	instrument-outcome association			Exposure Assumed Constant Over Pregnancy	
Lawlor	2008	Exploring the developmental overnutrition hypothesis using parental-offspring associations and FTO as an instrumental variable	point estimate	12.9, 10.1 after adjustment for offspring genotype		Pleiotropy, Postnatal Effects of Genotype, Offspring Genotype Path	Population Stratification
Lawlor	2017	Using Mendelian randomization to determine causal effects of maternal pregnancy (intrauterine) exposures on offspring outcomes: Sources of bias and methods for assessing them	point estimate	>45	Weak Instrument Bias	Pleiotropy, Postnatal Effects of Genotype, Offspring Genotype Path	Population Stratification, Assortative Mating

Lee	2013	Mendelian randomization analysis of the effect of maternal homocysteine during pregnancy, as represented by maternal MTHFR C677T genotype, on birth weight	point estimate			Pleiotropy, 2nd or 3rd Assumption - General	
Lewis	2009	Body composition at age 9 years, maternal folate intake during pregnancy and methyltetrahydrofolate reductase (MTHFR) C677T genotype	instrument-outcome association			Exposure Measurement Error	
Lewis	2012	Fetal Alcohol Exposure and IQ at Age 8: Evidence from a Population-Based Birth-Cohort Study	instrument-outcome association		Weak Instrument Bias	Pleiotropy	Population Stratification
Lewis	2014	Maternal iron levels early in pregnancy are not associated with offspring IQ score at age 8, findings from a Mendelian randomization study	instrument-outcome association			Pleiotropy,Offspring Genotype Path	Population Stratification
Mamasoula	2013	Association between C677T Polymorphism of Methylene Tetrahydrofolate reductase and Congenital Heart Disease: Meta-Analysis of 7697 Cases and 13,125 Controls. Circ Cardiovasc Genet	instrument-outcome association			2nd or 3rd Assumption - General	
Morales	2011	Maternal C-reactive protein levels in pregnancy are associated with wheezing and lower respiratory tract infections in the offspring	instrument-outcome association		Weak Instrument Bias	Pleiotropy, Exposure Assumed Constant Over Pregnancy,2nd or 3rd Assumption - General	
Morales	2016	Genome-wide DNA methylation study in human placenta identifies novel loci associated with maternal smoking during pregnancy	point estimate		Winner's Curse	Pleiotropy, Exposure Measurement Error,	

Murray	2016	Moderate alcohol drinking in pregnancy increases risk for children's persistent conduct problems: causal effects in a Mendelian randomisation study	instrument-outcome association		Cannot Prove Assumption 1 (Reduced Form)	Pleiotropy	Population Stratification
Richmond	2016	DNA methylation and BMI: Investigating identified methylation sites at HIF3A in a causal framework	point estimate	45.7 in offspring		Pleiotropy, Exposure Measurement Error, Offspring Genotype Path	Population Stratification
Richmond	2017	Using Genetic Variation to Explore the Causal Effect of Maternal Pregnancy Adiposity on Future Offspring Adiposity: A Mendelian Randomisation Study	point estimate	minimum 45	Weak Instrument Bias	Pleiotropy, Exposure Measurement Error, Postnatal Effects of Genotype, Offspring Genotype Path	Population Stratification
Scholder	2014	Alcohol Exposure In Utero and Child Academic Achievement	point estimate	1.38-24.76	Weak Instrument Bias	Pleiotropy, Postnatal Effects of Genotype, Exposure Assumed Constant Over Pregnancy,Offspring Genotype Path,2nd or 3rd Assumption - General	Population Stratification
Shaheen	2014	Prenatal alcohol exposure and childhood atopic disease: A Mendelian randomization approach	instrument-outcome association			Offspring Genotype Path	Population Stratification
Steenweg-de Graaff	2012	Maternal folate status in early pregnancy and child emotional and behavioral problems: the Generation R Study	instrument-outcome association			Exposure Assumed Constant Over Pregnancy,2nd or 3rd Assumption - General	Population Stratification
Steer	2011	Insights into the programming of bone development from the Avon Longitudinal Study of Parents and Children (ALSPAC)	instrument-outcome association			Offspring Genotype Path	

Taylor	2014	Maternal smoking during pregnancy and offspring smoking initiation: assessing the role of intrauterine exposure	instrument-outcome association			Pleiotropy, Exposure Measurement Error, Postnatal Effects of Genotype	Population Stratification
Thompson	2019	Association of maternal circulating 25(OH)D and calcium with birth weight: A mendelian randomisation analysis	point estimate		Weak Instrument Bias	Pleiotropy, Exposure Measurement Error, Exposure Assumed Constant Over Pregnancy, Offspring Genotype Path	Population Stratification
Tyrell	2016	Genetic evidence for causal relationships between maternal obesity related traits and birth weight. JAMA	point estimate		Weak Instrument Bias	Pleiotropy, Postnatal Effects of Genotype, Offspring Genotype Path	Population Stratification
Wehby	2011	A genetic instrumental variables analysis of the effects of prenatal smoking on birth weight: Evidence from two samples	point estimate	3.3-4.4	Weak Instrument Bias	Pleiotropy	Population Stratification
Wehby	2011	Genes as instruments for studying risk behavior effects: An application to maternal smoking and orofacial clefts	point estimate	3.33	Weak Instrument Bias	Pleiotropy, Exposure Measurement Error	
Wehby	2013	Genetic instrumental variable studies of effects of prenatal risk factors	point estimate	0.66-35.486	Weak Instrument Bias	Pleiotropy, Postnatal Effects of Genotype	Population Stratification
Yajnik	2014	Maternal homocysteine in pregnancy and offspring birthweight: Epidemiological associations and Mendelian randomization analysis	point estimate			Pleiotropy, Offspring Genotype Path	Population Stratification
Zerbo	2016	Maternal mid-pregnancy C-reactive protein and risk of autism spectrum disorders: the early markers for autism study	instrument-outcome association				Population Stratification

Zhang	2015	Assessing the Causal Relationship of Maternal Height on Birth Size and Gestational Age at Birth: A Mendelian Randomization Analysis	point estimate			Pleiotropy, Offspring Genotype Path	Assortative Mating
Zuccolo	2013	Prenatal alcohol exposure and offspring cognition and school performance. A 'Mendelian randomization' natural experiment	instrument-outcome association		Cannot Prove Assumption 1 (Reduced Form)	Pleiotropy, Exposure Measurement Error, Postnatal Effects of Genotype, 2nd or 3rd Assumption - General	Population Stratification

Table 1 Continued

1: Author	2: Year	3: Title	15: Other Limitations Discussed	16: Falsification Techniques	17:Exposure Stratification/Testing
Allard	2015	Mendelian randomization supports causality between maternal hyperglycemia and epigenetic regulation of leptin gene in newborns	Outcome Measurement Error, Low Power	Alternative MR Methods	No
Alwan	2012	Exploring the relationship between maternal iron status and offspring's blood pressure and adiposity: A Mendelian randomization Study.	Low Power	Covariate Balance	No
Bech	2006	Stillbirth and slow metabolizers of caffeine: Comparison by genotypes.	Low Power	Covariate Balance	Yes
Bedard	2018	Maternal iron status during pregnancy and respiratory and atopic outcomes in the offspring: A Mendelian randomisation study	Selection Bias, Low Power	Covariate Balance, Alternative MR Methods	No
Bernard	2018	Long-chain polyunsaturated fatty acids, gestation duration, and birth size: A Mendelian randomization study using fatty acid desaturase variants	Outcome Measurement Error, Low Power		No
Binder	2013	The causal effect of red blood cell folate on genome-wide methylation in cord blood: a Mendelian randomization approach	Modeling Assumptions, Outcome Measurement Error, Low Power,		No
Bonilla	2012	Maternal and offspring fasting glucose and type 2 diabetes-associated genetic variants and cognitive function at age 8: A Mendelian randomization study in the Avon Longitudinal Study of Parents and Children	Modeling Assumptions, Low Power	Covariate Balance	No

Bonilla	2012	Vitamin B-12 Status during Pregnancy and Child's IQ at Age 8: A Mendelian Randomization Study in the Avon Longitudinal Study of Parents and Children	,Selection Bias ,Low Power, Limited Generalizability	Covariate Balance	Yes
Caramaschi	2017	Exploring a causal role of DNA methylation in the relationship between maternal vitamin B12 during pregnancy and child's IQ at age 8, cognitive performance and educational attainment: a two-step Mendelian randomization study.	Low Power, Limited Generalizability		No
Caramaschi	2018	Maternal smoking during pregnancy and autism: using causal inference methods in a birth cohort study	Low Power	Covariate Balance	Yes
Evans	2018	Elucidating the role of maternal environmental exposures on offspring health and disease using two-sample Mendelian randomization	Modeling Assumptions	Alternative MR Methods	No
Geng	2018	Maternal central obesity and birth size: A Mendelian randomization analysis	Modeling Assumptions, Limited Generalizability	Alternative MR Methods	No
Granell	2008	The association between mother and child MTHFR C677T polymorphisms, dietary folate intake and childhood atopy in a population-based, longitudinal birth cohort	Modeling Assumptions, Selection Bias, Outcome Measurement Error	Covariate Balance	Yes
Howe	2019	Prenatal alcohol exposure and facial morphology in a UK cohort	Outcome Measurement Error	Covariate Balance	No
Humphriss	2013	Prenatal alcohol exposure and childhood balance ability: Findings from a UK birth cohort study	Outcome Measurement Error, Low Power, Limited Generalizability	Covariate Balance	No

Hwang	2019	Using a two-sample Mendelian randomization design to investigate a possible causal effect of maternal lipid concentrations on offspring birth weight	Limited Generalizability	Alternative MR Methods	No
Korevaar	2014	Soluble Flt1 and Placental Growth Factor are novel determinants of newborn thyroid dysfunction: the generation r study. J clin Endocrinol Metab	Low Power		No
Lawlor	2008	Exploring the developmental overnutrition hypothesis using parental-offspring associations and FTO as an instrumental variable	Selection Bias, Low Power	Covariate Balance	No
Lawlor	2017	Using Mendelian randomization to determine causal effects of maternal pregnancy (intrauterine) exposures on offspring outcomes: Sources of bias and methods for assessing them	Low Power	Covariate Balance, Alternative MR Methods	No
Lee	2013	Mendelian randomization analysis of the effect of maternal homocysteine during pregnancy, as represented by maternal MTHFR C677T genotype, on birth weight	Selection on Pregnancy, Low Power, Limited Generalizability	Covariate Balance	Yes
Lewis	2009	Body composition at age 9 years, maternal folate intake during pregnancy and methyltetrahydrofolate reductase (MTHFR) C677T genotype			No
Lewis	2012	Fetal Alcohol Exposure and IQ at Age 8: Evidence from a Population-Based Birth-Cohort Study	Selection Bias		Yes
Lewis	2014	Maternal iron levels early in pregnancy are not associated with offspring IQ score at age 8, findings from a Mendelian randomization study	Modeling Assumptions, Low Power	Covariate Balance	No
Mamasoula	2013	Association between C677T Polymorphism of Methylene Tetrahydrofolate reductase and Congenital Heart Disease: Meta-Analysis of 7697 Cases and 13,125 Controls. Circ Cardiovasc Genet	Modeling Assumptions		No
Morales	2011	Maternal C-reactive protein levels in pregnancy are associated with wheezing and lower respiratory tract infections in the offspring	Low Power		No

Morales	2016	Genome-wide DNA methylation study in human placenta identifies novel loci associated with maternal smoking during pregnancy	Low Power	Alternative MR Methods	No
Murray	2016	Moderate alcohol drinking in pregnancy increases risk for children's persistent conduct problems: causal effects in a Mendelian randomisation study	Selection Bias, Low Power	Covariate Balance	Yes
Richmond	2016	DNA methylation and BMI: Investigating identified methylation sites at HIF3A in a causal framework	Modeling Assumptions, Selection Bias, Low Power	Covariate Balance	No
Richmond	2017	Using Genetic Variation to Explore the Causal Effect of Maternal Pregnancy Adiposity on Future Offspring Adiposity: A Mendelian Randomisation Study	Modeling Assumptions, Selection Bias, Low Power	Covariate Balance, Alternative MR Methods	No
Scholder	2014	Alcohol Exposure In Utero and Child Academic Achievement	Modeling Assumptions	Weight Function, Covariate Balance	No
Shaheen	2014	Prenatal alcohol exposure and childhood atopic disease: A Mendelian randomization approach	Selection Bias, Low Power	Covariate Balance	Yes
Steenweg-de Graaff	2012	Maternal folate status in early pregnancy and child emotional and behavioral problems: the Generation R Study	Modeling Assumptions, Selection Bias		Yes
Steer	2011	Insights into the programming of bone development from the Avon Longitudinal Study of Parents and Children (ALSPAC)	NA		No
Taylor	2014	Maternal smoking during pregnancy and offspring smoking initiation: assessing the role of intrauterine exposure	Low Power, Limited Generalizability		Yes
Thompson	2019	Association of maternal circulating 25(OH)D and calcium with birth weight: A mendelian randomisation analysis	Outcome Measurement Error	Alternative MR Methods	No
Tyrell	2016	Genetic evidence for causal relationships between maternal obesity related traits and birth weight. JAMA	Modeling Assumptions, Low Power	Covariate Balance	No

Wehby	2011	A genetic instrumental variables analysis of the effects of prenatal smoking on birth weight: Evidence from two samples	Modeling Assumptions	Overidentification Tests	No
Wehby	2011	Genes as instruments for studying risk behavior effects: An application to maternal smoking and orofacial clefts	Modeling Assumptions	Overidentification Tests	No
Wehby	2013	Genetic instrumental variable studies of effects of prenatal risk factors	Modeling Assumptions, Low Power, Limited Generalizability	Covariate Balance	No
Yajnik	2014	Maternal homocysteine in pregnancy and offspring birthweight: Epidemiological associations and Mendelian randomization analysis	Low Power		No
Zerbo	2016	Maternal mid-pregnancy C-reactive protein and risk of autism spectrum disorders: the early markers for autism study	Outcome Measurement Error, Low Power		No
Zhang	2015	Assessing the Causal Relationship of Maternal Height on Birth Size and Gestational Age at Birth: A Mendelian Randomization Analysis	Selection Bias	Alternative MR Methods	No
Zuccolo	2013	Prenatal alcohol exposure and offspring cognition and school performance. A 'Mendelian randomization' natural experiment	Low Power	Covariate Balance	Yes

IV. Assumptions required for point estimation

Investigators can test whether there is a non-null effect of the exposure on the outcome for at least one individual in the study population, and can estimate bounds for the average causal effect using only the 3 instrumental variable assumptions discussed in the main text [1, 2].

To estimate the average causal effect of an exposure X on an outcome Y, using an instrument Z in the total study population, investigators must assume one of the following conditions hold [2].

4a. The effect of X on Y is identical (constant) for all individuals in the population:

$E(Y^{x=x}-Y^{x=0} \mid X=x)=E(Y^{x=x}-Y^{x=0} \mid X=0)$

4b. No effect modification by the instrument Z in all levels of the exposure X:

$E(Y^{x=x}-Y^{x=0} \mid X=x, Z=z)=E(Y^{x=x}-Y^{x=0} \mid X=x, Z=\tilde{z})$

Or equivalently

$E(Y^{x=x}-Y^{x=0} \mid X=x, Z=z)=E(Y^{x=x}-Y^{x=0} \mid X=\tilde{x}, Z=z)$

Recent research has found that the average causal effect can also be identified, even in the presence of violations of the second and third assumption, under one of two alternative assumptions by an additional variable J, as can be seen in Supplemental Figure 1 [3]. In this case, the usual 3 IV assumptions are replaced by the following:

1'. $Z \not\perp\!\!\!\perp X \mid J$

2'. $Z \perp\!\!\!\perp U \mid X$

3'. $Z \perp\!\!\!\perp Y \mid (J, U, X)$

4'. $Y^{x} \perp\!\!\!\perp (Z, X) \mid (J, U)$

Under these conditions, point estimation of the average causal effect is possible if one of the two following conditions hold:

4c. No additive U-Z interaction on E(X | Z, J, U):

$$E(X \mid Z=z, J, U) - E(X \mid Z=0, J, U) = E(X \mid Z=z, J) - E(X \mid Z=0, J)$$

4d. No additive U-X interaction on the average causal effect of X on Y:

$$E(Y^{X=x} - Y^{X=0} \mid J, U) = E(Y^{X=x} - Y^{X=0} \mid J)$$

If the above assumptions are not plausible for a particular analysis, researchers can estimate the average causal effect within the compliers, those individuals for whom $X^{Z=a} > X^{Z=b}\ for\ all\ a > b$ [2]. This value is also known as the local average treatment effect, or LATE. In order to estimate this quantity, researchers must assume:

4e. The causal effect of Z on X is monotonic, that is, it only works in one direction for every individual in the study population. Formally, X^z is a nondecreasing function of z on the support of Z .

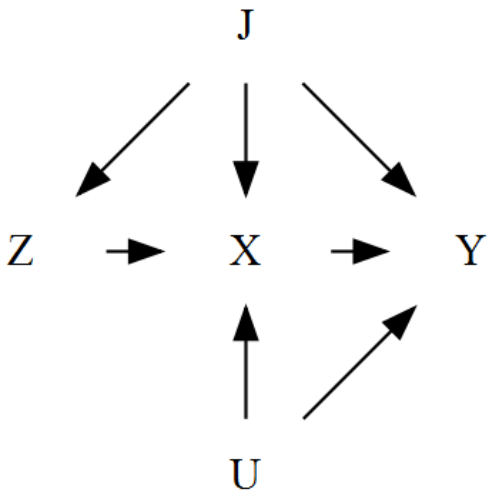
V. Interpretation of certain additional point-estimating assumptions in pregnancy Mendelian randomization

Four studies in this review reported additional point-estimating assumptions and their targeted estimand. Of these, 3 assumed monotonicity (assumption 4e) in order to estimate the average causal effect among the compliers, and 1 assumed no effect modification by the instrument in all levels of the exposure (assumption 4b) to estimate the average causal effect in the total study population.

In the context of certain pregnancy exposures, there is evidence that conditions 4a, 4b, 4c, and 4d are unreasonable. When genes related to alcohol metabolism are used as instruments for maternal drinking during pregnancy, fetal exposure to alcohol and alcohol metabolites will depend on maternal intake and the speed at which the mother can metabolize alcohol, as well as other environmental factors. For the same level of maternal alcohol intake, offspring of slow metabolizers will have a longer exposure to alcohol, and would be at greater risk of negative health outcomes [5]. This means that the average causal effect of alcohol exposure on offspring outcomes will be modified by the level of the maternal genetic variant proposed as an instrument, violating conditions 4a, 4b, 4c, and 4d. For this reason, most studies of alcohol use during pregnancy in this review focused on a testing approach, rather than point estimation. The same logic applies to other metabolism-related genetic variants proposed as instruments for substance use behaviors, like smoking and caffeine use. In these cases, studies may choose to focus on approaches with weaker assumptions, such as the complier average treatment effect, testing approaches, or bounds.

It is important to note that, in prenatal MR proposing maternal genetic factors as instruments, the interpretation of “compliers” and the complier average causal effect (described in Appendix III above) are different than the usual interpretation in MR studies or most studies using instrumental variable analyses[4]. This is because a mother-child pair’s compliance status is determined by the relationship between a mother’s genetics and exposure, while the average causal effect of interest occurs in the offspring of those mothers. In typical MR and instrumental variable studies, the proposed instrument, exposure, and outcome are all measured within the same individual. In those cases, under condition 4e, researchers can estimate the average causal effect among the compliers. In contrast, in pregnancy MR designs, under condition 4e, researchers can estimate the average

causal effect among the offspring of mothers who are compliers, although the offspring themselves would not necessarily be compliers.



Supplemental Figure 1: DAG representing an instrumental variable model with violation of assumption 3 by J. Under this model, valid estimation of $E(Y^{X=x} - Y^{X=0})$ is possible using the alternative assumptions presented by Wang and Tchetgen Tchetgen [3].

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